Supplementary Material S11: Judging AI annotations against manual annotation, an example of five protocols

We manually annotated five protocols and compared them with annotations produced by large language models. Each AI annotation is scored against the manual annotation of the same entry. Table S11-1 describes how each model was accessed and the sampling parameters used; Table S11-2 gives the F1, precision and recall of every model and route.

1. Data and metric

Five eLabFTW entries were used: bacteriophage sequencing, spot assay, strain generation, BioLector cultivation and plasmid generation. A curator manually annotated each eLabFTW entry. The metrics include record-level precision, recall and F1. Rows are paired one-to-one across all tables of the assay, and the cells of each pair are then compared after normalising their format (decimal comma, units, dates, case, "~"). A wrong, missing or invented row counts as an error. A cell is a non-empty value cell of the assay tables (columns Characteristic, Parameter, Factor, Comment and Date), counted per row; a value and its unit count once. A failed run (invalid or truncated JSON, no usable answer) scores 0 over the full annotation. An endpoint error (rate limit, quota, server down) is not a model failure; such runs were repeated and never scored. All numbers in the tables were computed by program from the per-run scores.

2. Task and routes

We tested more than 10 prompt versions, all published in the GitHub repository. Of these, version 6, a single-call structured-extraction prompt, scored highest on average. The model receives the text of the entry as the application converts it (HTML to markdown), without figures (adding figure descriptions or the images themselves did not improve the scores (Section 3)). One request is sent per run, and the answer is parsed as strict JSON without repair. Table S11-1 lists how each model was accessed and its sampling parameters, which follow the model cards. Commercial models (Claude Opus 5.5 and Sonnet 5.5) were used through their command-line interface in print mode, Kimi-K3 through the Kimi Code CLI, an agent harness, and DeepSeek-V4.1-Flash through a hosted API. Open-weight models were run locally on a Mac (LM Studio with 4- to 6-bit builds, oMLX with 4-bit), on the institute GPU cluster (vLLM) and through free-tier OpenRouter endpoints. Because quantisation and route differ between rows, comparisons across routes are approximate. Table S11-2 gives the scores of all models and routes, where n is the number of repeats per cell.

Table S11-1. Access route and sampling parameters of the models

Model and access route	Sampling parameters
Claude Opus 5.5, Sonnet 5.5 (CLI, print mode)	CLI defaults; no sampling parameters set by the runner
Kimi-K3 (Kimi Code CLI, agent harness)	harness defaults; no sampling parameters set by the runner
DeepSeek-V4.1-Flash (hosted API)	the application's own settings: temperature 1.0, top_p 0.95, repetition_penalty 1.1, thinking off, max_tokens 28000
Gemma-4-31B, GLM-4.7-Flash (LM Studio)	Gemma: temperature 1.0, top_p 0.95, top_k 64, max_tokens 32768; GLM: temperature 1.0, top_p 0.95, max_tokens 65536
MiniMax-M3 (oMLX)	temperature 1.0, top_p 0.95, max_tokens 65536
Gemma-4-31B (GPU cluster, vLLM)	temperature 1.0, top_p 0.95, top_k 64, max_tokens 32768
OpenRouter free-tier models	temperature 0, max_tokens 64000

Table S11-2. Extraction prompt (version 6) against the manual annotation: F1 and precision / recall

Model	Route	Bacteriophage sequencing	Spot assay	Strain generation	BioLector cultivation	Plasmid generation	Mean F1	Mean P	Mean R	Entries	Runs	Failed
Claude Opus 5.5	Claude CLI, print mode	**0.61 (1)** 0.53 / 0.71	**0.72 (1)** 0.71 / 0.73	**0.49 (1)** 0.74 / 0.37	**0.48 (1)** 0.47 / 0.50	**0.65 (1)** 0.75 / 0.57	**0.59**	0.64	**0.58**	5	5	0
Claude Sonnet 5.5	Claude CLI, print mode	**0.58 ± 0.01 (3)** 0.51 / 0.67	**0.42 ± 0.13 (3)** 0.58 / 0.34	**0.50 ± 0.03 (3)** 0.78 / 0.37	**0.23 ± 0.03 (3)** 0.17 / 0.36	**0.37 ± 0.01 (3)** 0.63 / 0.26	0.42	0.53	0.40	5	15	0
dots-3-note-preview	OpenRouter free tier	**0.62 (1)** 0.63 / 0.60	**0.33 (1)** 0.31 / 0.35	**0.25 (1)** 0.41 / 0.18	**0.22 (1)** 0.17 / 0.29	**0.45 (1)** 0.54 / 0.38	0.37	0.41	0.36	5	5	0
Gemma-4-31B	Mac, LM Studio, GGUF Q4_K_M	**0.63 ± 0.07 (3)** 0.74 / 0.55	**0.41 ± 0.07 (3)** 0.76 / 0.28	**0.16 ± 0.02 (3)** 0.71 / 0.09	**0.27 ± 0.02 (3)** 0.23 / 0.31	**0.21 ± 0.04 (3)** 0.72 / 0.12	0.33	0.64	0.27	5	15	0
Gemma-4-31B	GPU cluster, vLLM, QAT w4a16	**0.56 ± 0.08 (3)** 0.78 / 0.43	**0.36 ± 0.00 (3)** 0.88 / 0.22	**0.12 ± 0.05 (3)** 0.72 / 0.06	**0.43 ± 0.01 (3)** 0.57 / 0.34	**0.19 ± 0.07 (3)** 0.66 / 0.11	0.33	**0.72**	0.24	5	15	0
Nemotron-3-Ultra-550B	OpenRouter free tier	**0.41 (1)** 0.37 / 0.45	**0.47 (1)** 0.50 / 0.44	**0.31 (1)** 0.66 / 0.20	**0.00 (1)*** 0.00 / 0.00	**0.36 (1)** 0.32 / 0.42	0.31	0.37	0.30	5	5	1
MiniMax-M3-4bit	Mac, oMLX, 4-bit	**0.60 ± 0.05 (3)** 0.51 / 0.75	**0.36 ± 0.32 (3)*** 0.37 / 0.35	**0.12 ± 0.20 (3)*** 0.17 / 0.09	**0.08 ± 0.13 (3)*** 0.06 / 0.12	**0.35 ± 0.01 (3)** 0.53 / 0.26	0.30	0.33	0.31	5	15	5
DeepSeek-V4.1-Flash	hosted API	**0.53 ± 0.07 (3)** 0.44 / 0.66	**0.18 ± 0.13 (3)** 0.39 / 0.14	**0.12 ± 0.21 (3)*** 0.23 / 0.08	**0.25 ± 0.03 (3)** 0.21 / 0.32	**0.32 ± 0.00 (3)** 0.50 / 0.24	0.28	0.36	0.29	5	15	2
GLM-4.7-Flash	Mac, LM Studio, MLX 6-bit	**0.33 ± 0.03 (3)** 0.41 / 0.28	**0.09 ± 0.12 (3)*** 0.28 / 0.06	**0.25 ± 0.02 (3)** 0.46 / 0.21	**0.31 ± 0.05 (3)** 0.39 / 0.26	**0.32 ± 0.09 (3)** 0.41 / 0.32	0.26	0.39	0.23	5	15	1
Kimi-K3	Kimi Code CLI (agent harness)	**0.60 ± 0.02 (3)** 0.54 / 0.67	**0.04 ± 0.07 (3)*** 0.11 / 0.02	**0.00 ± 0.00 (3)*** 0.00 / 0.00	**0.28 ± 0.08 (3)** 0.25 / 0.34	**0.30 ± 0.28 (3)*** 0.41 / 0.24	0.24	0.26	0.26	5	15	6

*First line of each cell: F1 as mean ± sd over the n runs in brackets (no sd for a single run); * marks a mean that includes a failed run, scored as 0. Second line: mean precision / mean recall, with failed runs as 0. Mean F1, P and R are the means of the five entry values. Rows are sorted by mean F1, and only models run on all five entries are shown. Models and routes are described in Table S11-1. Source: per-run scores, prompt version 6, text-only input.*

3. Additional material and reference annotation

A second manual annotation, made by the study authors, exists for one of the entries. Their precision, recall and F1 against each other are provided in the GitHub repository https://github.com/usadellab/protocol-eva and serve as a reference for interpreting the precision, recall and F1 of the AI results. We also tested prompts that requested verbatim evidence quotes (traceable extraction) and prompts that described figures, either as descriptions written by a vision model or as the images themselves. Neither improved the average scores, testing resources were limited, and they are not the focus of the elab2arc publication. Therefore, other scores are not part of this supplement. They are documented in the repository (folders “benchmarking_versions”, “traceable” and “image_addition”).